GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-031 Entry to Kingsmead Cable Tunnel (Confined Space)

Field	Value
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Owner	Head of Electricity Network Operations
Approved by	Electricity Safety Manager
Applies to	Kingsmead Primary Substation (ESS-KGM): Kingsmead cable tunnel (CT-KGM-01)

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how people enter, work in and leave the Kingsmead cable tunnel (CT-KGM-01) safely, and how they are rescued if something goes wrong.

2 Scope

CT-KGM-01 is a 420 m cable tunnel at Kingsmead Primary Substation (ESS-KGM). It carries the 33 kV cables supplying T-201 and T-202 and 11 kV feeder cables including FDR-KGM-03. Shaft A, inside the ESS-KGM compound, is the only entry point. Shaft B, at the far end, is an emergency exit only.

The tunnel is a confined space under the Confined Spaces Regulations 1997 because of the foreseeable risks of oxygen deficiency, toxic or flammable gas entering from the ground or nearby sewers, smoke from a cable fault and flooding. This procedure applies to every entry, whether for inspection or for work.

3 References

- PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff
- PR-ELEC-012 Isolation and Earthing of 11 kV Circuits
- PR-ELEC-015 Switching Programmes on the 11 kV and 33 kV Networks
- PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- HSE HSG250 Guidance on permit-to-work systems

- Confined Spaces Regulations 1997
- Management of Health and Safety at Work Regulations 1999
- Electricity at Work Regulations 1989

4 Definitions

Confined space entry permit: a permit issued by an Authorised Person (HV) allowing a stated team to enter CT-KGM-01 for a stated task and period.

Entrant: a person who enters the tunnel.

Top Person: the person stationed at the head of Shaft A for the whole entry, sometimes called the top man. The Top Person never enters the tunnel.

Rescue team: two people trained in confined space rescue and the use of self-contained breathing apparatus, on standby at ESS-KGM during every entry.

Escape set: a self-contained escape breathing apparatus giving at least 15 minutes of air.

Gas monitor: an approved four-gas monitor measuring oxygen, flammable gas (as a percentage of the lower explosive limit, LEL), carbon monoxide and hydrogen sulphide.

5 Responsibilities

Head of Electricity Network Operations: provides the rescue arrangements and makes sure the rescue plan is exercised at least once a year.

Senior Authorised Person (HV): approves the task risk assessment for any entry other than routine inspection.

Authorised Person (HV): issues and cancels the confined space entry permit, and confirms atmosphere tests, ventilation, communications and rescue cover before anyone enters.

Competent Person (HV) acting as Top Person: controls the entry log, keeps contact with entrants and starts the rescue plan.

Competent Person (HV) acting as entrant: follows the permit, monitors the atmosphere and leaves when told to or when an alarm sounds.

Control Engineer: records the entry, flags the tunnel circuits on SCADA, and does not re-energise a tunnel circuit that trips while anyone is inside.

6 Procedure

6.1 Planning the entry

6.1.1 Decide whether the task can be done without entering the tunnel. Enter only if it cannot.

6.1.2 Prepare a task risk assessment. For any entry other than routine inspection, obtain the approval of the Senior Authorised Person (HV).

6.1.3 Arrange a team of at least two entrants, a Top Person and a two-person rescue team, all with current confined space training. Lone entry is prohibited.

6.1.4 Check the flood warnings for the Kingsmead area and the tunnel sump level alarm. Do not enter if a flood warning is in force or the sump high-level alarm is active.

6.1.5 Inform the Control Engineer of the planned entry, who places a "persons in tunnel" flag on every circuit in CT-KGM-01 on SCADA.

6.1.6 As Authorised Person (HV), issue the confined space entry permit stating the task, the number of entrants, the maximum duration, the communication method and the rescue arrangements.

6.1.7 Where the task involves work on a tunnel cable, also isolate it under PR-ELEC-012 and issue a Permit-to-Work under PR-ELEC-020.

6.2 Ventilation

6.2.1 Open the covers of Shaft A and Shaft B with the lifting equipment provided, and place barriers around both openings.

6.2.2 Start the fixed supply fan at Shaft A and confirm that the airflow indicator at Shaft B shows air leaving the tunnel.

> CAUTION: Exhaust from vehicles or generators drawn into the fan intake can fill the tunnel with carbon monoxide.

6.2.3 If the fixed fan has failed, use the portable fan and ducting from the ESS-KGM store, with the intake upwind and at least 10 m from any vehicle or generator.

6.2.4 Run forced ventilation for at least 30 minutes before the pre-entry atmosphere test.

6.2.5 Keep forced ventilation running for the whole time anyone is in the tunnel.

> WARNING: Without forced ventilation, oxygen can fall and gas can build up quickly in the far end of the tunnel.

6.2.6 As Top Person, if the ventilation stops during an entry, order all entrants out immediately.

6.3 Atmosphere testing

Gas	Alarm level
Oxygen	below 19.5% or above 23.0% by volume
Flammable gas	10% LEL
Carbon monoxide	20 ppm
Hydrogen sulphide	5 ppm

6.3.1 Bump test every gas monitor with test gas before use, and check it has been calibrated within the last six months.

6.3.2 From the head of Shaft A, sample the atmosphere through the sampling line at the top, middle and bottom of the shaft.

6.3.3 Repeat the test at the head of Shaft B and record all readings on the entry permit.

> HOLD POINT: Nobody may enter the tunnel until the Authorised Person (HV) has checked that every pre-entry reading is within the alarm levels and has signed the entry permit.

6.3.4 Each entrant wears a personal gas monitor, switched on, for the whole entry.

6.3.5 Advance in stages, checking the monitor display at least every 50 m.

6.3.6 If any monitor alarms, leave the tunnel immediately by the nearest exit. Do not re-enter until the Authorised Person (HV) has found the cause and the tunnel has been retested.

6.4 Entry and communications

6.4.1 Before entry, check that each entrant has a hard hat with lamp, safety footwear, overalls, an escape set, a personal gas monitor and a radio.

6.4.2 As Top Person, record each entrant's time in on the entry log board.

6.4.3 Test the tunnel leaky feeder radio with the Top Person at the foot of Shaft A and again at 100 m before going further.

6.4.4 As entrant, report to the Top Person at least every 15 minutes, giving your chainage marker and gas readings.

6.4.5 As Top Person, if a check-in is missed and contact cannot be made within 5 minutes, start the rescue plan in 6.7.

6.4.6 As Top Person, stay at the head of Shaft A for the whole entry and take on no other task.

> WARNING: Cables in the tunnel are live unless covered by a Permit-to-Work. A cable fault can cause an explosion and fire, and damaged cable can be lethal to touch.

6.4.7 Do not touch, stand on or move cables, joints or cable cleats unless the work is covered by a Permit-to-Work.

6.5 Maximum entry durations

6.5.1 Do not stay in the tunnel for more than 2 hours at a time. Spend at least 30 minutes at the surface after each period.

6.5.2 Do not spend more than 4 hours in total in the tunnel in one shift.

6.5.3 Allow no more than four entrants in the tunnel at once.

6.5.4 Where the tunnel air temperature is above 30 °C, limit each period to 1 hour.

6.5.5 Treat the entry permit as expired at the end of the shift or after 10 hours, whichever comes first.

6.6 Leaving the tunnel and closing the permit

6.6.1 As entrant, report to the Top Person on leaving the shaft.

6.6.2 As Top Person, record each time out and confirm that every entrant on the log board is out.

6.6.3 As Authorised Person (HV), confirm the count, cancel the entry permit and inform the Control Engineer, who removes the SCADA flag.

6.6.4 Stop the ventilation, replace and lock both shaft covers.

6.6.5 Report any defects found, such as water ingress, damaged cable supports or failed lighting, to the Control Engineer.

6.7 Rescue and emergency

> WARNING: People who go into a confined space to rescue someone without breathing apparatus often become casualties themselves. The Top Person must never enter the tunnel.

6.7.1 As Top Person, on any gas alarm, missed check-in or casualty, order all entrants out and call 999. Ask for the fire and rescue service and an ambulance to a confined space rescue at Kingsmead Primary Substation.

6.7.2 Inform the Control Engineer, who de-energises tunnel circuits by fault switching under PR-ELEC-015 if the rescue team leader asks.

6.7.3 Send in the site rescue team only wearing self-contained breathing apparatus, following the ESS-KGM rescue plan.

6.7.4 Meet the emergency services at the ESS-KGM gate with the rescue plan and tunnel drawing, and guide them to Shaft A.

6.7.5 In the event of smoke or fire, leave by the nearest shaft and do not attempt to fight the fire.

6.7.6 Ask members of the public who report a power cut to call 105, and report the event under PR-CORP-001.

7 Records

- Confined space entry permits and entry log sheets, kept for six years.
- Gas monitor calibration and bump test records.
- Rescue plan exercise records, kept by the Head of Electricity Network Operations.

8 Revision history

Version	Date	Change
1	2023-11-06	First issue, replacing the general confined space instruction for ESS-KGM.
2	2025-11-03	Added hydrogen sulphide to the atmosphere tests. Set maximum entry durations, including the reduced limit above 30 °C. Required a two-person rescue team on site for every entry. Added 15-minute check-ins on the leaky feeder radio and the SCADA "persons in tunnel" flag.